

Course Syllabus

Department of Mechanical Engineering
Chulalongkorn University

Academic Year: Fall 2026 (First Semester)

Course Information

Course Number	2103304
Course Title	Automatic Control I
Credits	3 (3–0–9)
Semester	Fall 2026 (First Semester)
Instructor	Chawin Ophaswongse, Ph.D.
Co-Instructor	Assoc. Prof. Thitima Jintanawan, Ph.D.
Class Time	Monday, 13:00–16:00
Location	Room 409, Building 3, Faculty of Engineering

Course Description

Official course description.

Introduction to control systems; mathematical models of systems; feedback control system characteristics; the performance of feedback control systems; the stability of linear feedback systems; essential principles of feedback; the root locus method; frequency response method; stability of the frequency domain, time-domain analysis of control systems: the design and compensation of feedback control system.

Scope of the course. Control engineering is concerned with making a physical system behave in a prescribed manner, typically faster, more stable, and more accurate than it would be without control. This course develops the complete workflow, from a physical system to a validated controller:

1. **Modeling.** Represent a dynamic system of any physical domain, whether mechanical, electrical, electro-mechanical, thermal, fluid, or a combination of these, by differential equations, transfer functions, and state-space models, derived either from first principles or fitted to measured data (*system identification*).
2. **Analysis.** Determine how the system responds in the time and frequency domains, including transient response, steady-state error, and stability.
3. **Design.** Select and tune a controller (P, PI, PD, PID, lead/lag, or full-state feedback) so that the closed-loop system satisfies specifications such as rise time, overshoot, and settling time.

Software and hardware. All techniques are practiced in **MATLAB/Simulink** with the Control System Toolbox. The *Servo Mini-labs* and the *Final Project* extend this work to a physical servo plant (a brushed DC

motor), which students identify, simulate, control, and tune. All interaction with the hardware takes place through MATLAB/Simulink; **no low-level or embedded programming is required**, so that coursework remains focused on control-engineering decisions rather than firmware implementation.

Course Objectives

Upon successful completion of this course, students will be able to:

- **Derive the governing equations** of a mechanical or electro-mechanical system and establish a systematic procedure for analyzing a real engineering problem.
- **Construct engineering models** (free-body diagrams, kinematics and kinetics diagrams, and control block diagrams) and translate them into differential-equation, transfer-function, and state-space representations.
- **Analyze system behavior** in the time and frequency domains, including transient response, steady-state error, and stability.
- **Design and tune feedback controllers** (P/PI/PD/PID, lead/lag, and full-state feedback) to meet performance specifications, using root-locus and frequency-response methods.
- **Apply MATLAB/Simulink** to identify, model, simulate, analyze, and design control systems, and validate a controller on a physical servo plant.

References

Primary

- Nise, N. S. (2011). *Control Systems Engineering* (6th ed.). Wiley.
- Lecture and Handout Notes (provided on MyCourseVille)

Supplementary

- Ogata, K. (2010). *Modern Control Engineering* (5th ed.). Prentice Hall.
(*State space; MATLAB; more mathematically rigorous*)
- Dorf, R. C., & Bishop, R. H. (2017). *Modern Control Systems* (13th ed.). Pearson.
(*Design emphasis; rich in illustrative examples*)

Course Schedule

Wk	Date	Topic	Key Content	HW	Servo Mini-lab
1	3 Aug	Intro & Math Primer	Open-/closed-loop · ODEs · LTI · Laplace Transform	HW1	
2	10 Aug	System Modeling I	Transfer functions · Electrical/mechanical systems		Lab 1: Intro & Setup
3	17 Aug	System Modeling II	State space · Nonlinearities · Linearization	HW2	
4	24 Aug	Time Response	1st/2nd-order · Transient-response specifications	HW3	Lab 2: Time Response
5	31 Aug	Block Diagrams	Diagram Reduction · Equivalent forms · Feedback		
6	7 Sep	Stability	Routh–Hurwitz · Gain range for stability	HW4	Lab 3: P-Control & Stability
7	14 Sep	Midterm Review	Review of Weeks 1–6		

Midterm Examination: Wednesday, 23 September 2026, 13:00–16:00

8	28 Sep	Steady-State Errors	System type · Gain design · Disturbances	HW5	
9	5 Oct	Root Locus I	Sketching rules · Transient response via gain	HW6	
10	12 Oct	Root Locus II	PI/PD/PID · Lead/lag compensation		Lab 4: Root Locus
11	19 Oct	Frequency Response I	Asymptotic Bode plots · Bandwidth	HW7	
12	26 Oct	Frequency Response II	Nyquist & margins · Lead/lag · PID		Lab 5: Frequency Response
13	2 Nov	Design via State Space	Controllability · Full-state feedback	HW8	Lab 6: Pole Placement
14	9 Nov	Project Demo Day	Group presentations		
15	16 Nov	Final Review	Review of all topics		

Final Examination: Monday, 30 November 2026, 08:30–11:30

Course Policies

- **Communication.** Announcements and course materials are posted on MyCourseVille.
- **Attendance.** Attendance at mini-lab sessions is **mandatory and graded**.
- **Mini-labs.** Pre-lab handouts are released one week before each session and constitute required reading. Lab reports (groups of up to four) are due **one week** after the session.
- **Assignments.** Homework is due **ten days** after it is assigned unless otherwise specified. Late submissions are penalized according to the stated policy.
- **Academic integrity.** Collaboration is encouraged; however, all submitted work must be the individual's or the group's own. Plagiarism and examination misconduct are handled under Faculty of Engineering regulations.
- **Software.** MATLAB/Simulink and the Control System Toolbox must be installed before Week 2.

Assessment

Component	Weight
Assignments (Homework)	10%
Servo Mini-labs	10%
Midterm Examination	30%
Final Project	15%
Final Examination	35%
Total	100%

Servo Mini-labs

Six 45-minute hands-on sessions are conducted at the end of the lectures. Students identify a plant model experimentally (system identification), model its components (actuator, sensor, and controller), and then design and tune controllers in **MATLAB/Simulink** for validation on a real-time servo setup consisting of a brushed DC motor with an interchangeable inertia-disk or pendulum load. From Lab 2 onward, each session compares the no-load and inertia-disk conditions, so that the effect of an added load on the same system can be observed directly.

Requirements. A laptop with MATLAB/Simulink and the Control System Toolbox is required. Sessions are mandatory and graded. All interaction with the hardware takes place through the Simulink control interface; *no low-level or embedded programming is required.*

- **Lab 1: Introduction & Setup (Wk 2).** Connect the DC-motor and inertia-disk plant to Simulink; become familiar with the control interface, encoder sensing, and actuation; conduct open-loop step tests and an initial system identification of the motor parameters.
- **Lab 2: Time Response (Wk 4).** Measure open- and closed-loop step responses; extract first- and second-order parameters together with the transient specifications (rise time, overshoot, and settling time) and compare them against the identified model, with and without the inertia disk.
- **Lab 3: P-Control & Stability (Wk 6).** Implement proportional position and speed control; observe how the gain shapes the transient response and the onset of instability, and how the added inertia changes both.
- **Lab 4: Root Locus (Wk 10).** Design the controller gain (P/PD/PID) by root locus to meet transient specifications and validate the design on the plant under both load conditions.
- **Lab 5: Frequency Response (Wk 12).** Measure the frequency response of the plant (Bode) with and without the inertia disk; evaluate the gain and phase margins and tune a lead/lag or PID controller.
- **Lab 6: Pole Placement (Wk 13).** Design full-state feedback by pole placement for the DC motor with a load and validate the closed-loop response.

Final Project

Beginning after the midterm, students work in groups of 3–4 to carry out the complete workflow on an assigned electro-mechanical plant: performing system identification to obtain the plant model, constructing a **Simulink simulation model** and validating it against the measured hardware response, and then designing, tuning, and demonstrating a controller that meets the target performance specifications. The comparison between simulation and hardware is a central theme; students are expected to explain and reconcile the discrepancies they observe.

Tentative milestones (subject to adjustment):

- **Milestone 1: System Identification & Simulation Model (Week 9, 5 Oct).** Conduct experiments to identify the plant parameters; construct a Simulink simulation model and validate it against the measured open-loop hardware response.
- **Milestone 2: Controller Design in Simulation (Week 11, 19 Oct).** Specify the target performance, design and tune the controller in simulation, and verify that the specifications are met on the simulation model.
- **Milestone 3: Hardware Implementation & Tuning (Week 13, 2 Nov).** Deploy the controller on the physical plant, retune as required, and document the discrepancies between simulation and hardware.
- **Final Deliverable: Demo Day & Report (Week 14, 9 Nov).** Present the final closed-loop performance at the demo-day and submit the final project report.

This syllabus is subject to minor adjustment; any changes will be announced in class and on MyCourseVille.